



CASE STUDY PROBLEM: Electrical–Mechanical Analogues Using RLC and MSD Systems

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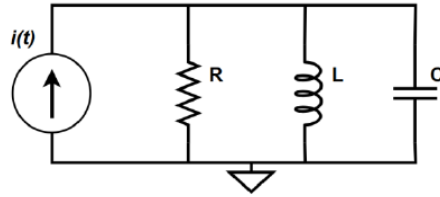
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Part 1: Electrical to Mechanical (Force-Current) Conversion

Objective: You are responsible for analyzing and converting the following electrical circuit to an analogue mechanical system.



Task 1 (EE): Write the node equation for the show circuit above.

- According to Kirchhoff's Current Law (KCL), total current passing through the circuit is the total sum of the currents passing through the resistor, inductor, and capacitor.

$$i(t) = i_R + i_L + i_c$$

If we write this equation in terms of voltage($V(t)$) we get the equation:

$$i(t) = \frac{V(t)}{R} + \frac{1}{L} \int_0^t V(\tau) d\tau + C \frac{dV(t)}{dt}$$

($V(t)$ is voltage across the capacitor)

Task 2 (EE): Perform an LTSpice simulation to visualize its unit step response. (*Delivered as a separate LTspice file*)

Task 3 (ME): Rewrite the equations using mechanical analogies.

The provided electrical system is a parallel RLC circuit driven by a current source. Applying Kirchhoff's Current Law (KCL) yields the following node equation:

$$C \frac{dV}{dt} + \frac{V}{R} + \frac{1}{L} \int V dt = i(t)$$

To convert this to an equivalent mechanical system, we apply the Force-Current (F-I) Analogy:

Force corresponds to current: $f(t) = i(t)$

Velocity corresponds to voltage: $v = V$

Mass corresponds to capacitance: $m = C$

Damping coefficient corresponds to inverse resistance: $b = \frac{1}{R}$

Spring stiffness corresponds to inverse inductance: $k = \frac{1}{L}$

Substituting these mechanical analogues into the KCL equation, we obtain the differential equation for the mechanical system in terms of velocity (v) and displacement (x):

$$m \frac{dv}{dt} + bv + k \int v dt = f(t)$$

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = f(t)$$

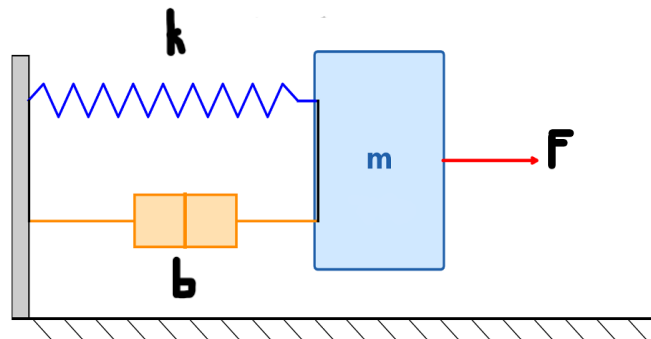
Task 4 (ME): Draw the mechanical system that corresponds to the differential equation and state its analytical solution.

Since the electrical components share the same voltage across parallel branches, the analogous mechanical elements must share the same velocity. Therefore, the equivalent mechanical design consists of a mass (m), a spring (k), and a damper (b) connected in parallel to a fixed reference frame, all subjected to the external force $f(t)$.

For a unit step response (constant force F_0 applied at $t = 0$), the analytical solution for the displacement $x(t)$ of an underdamped system ($0 < \zeta < 1$) is given by:

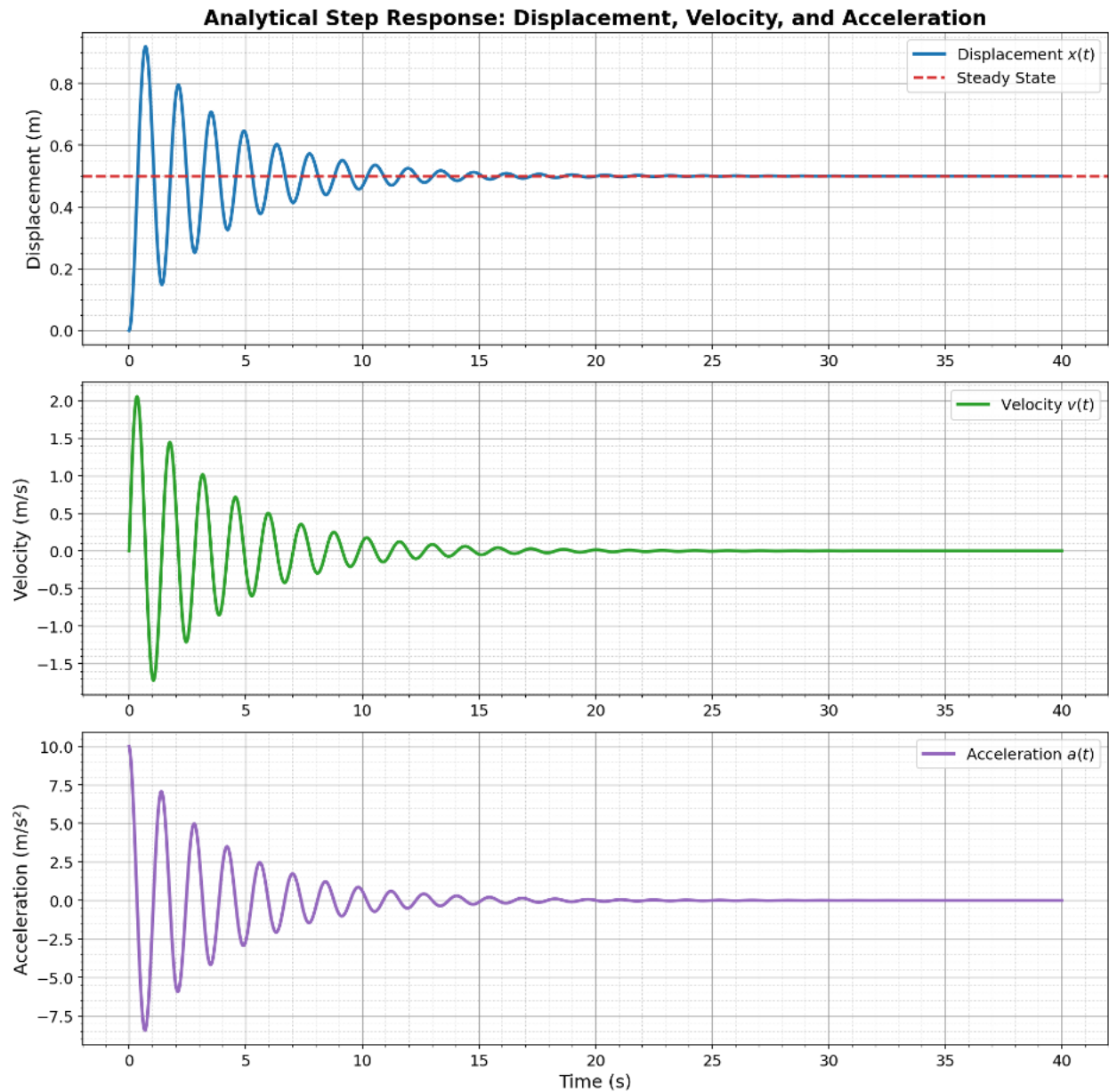
$$x(t) = \frac{F_0}{k} \left[1 - e^{-\zeta \omega_n t} \left(\cos(\omega_d t) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(\omega_d t) \right) \right]$$

Where $\omega_n = \sqrt{\frac{k}{m}}$ is the natural frequency, $\zeta = \frac{b}{2\sqrt{mk}}$ is the damping ratio, and $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ is the damped natural frequency.

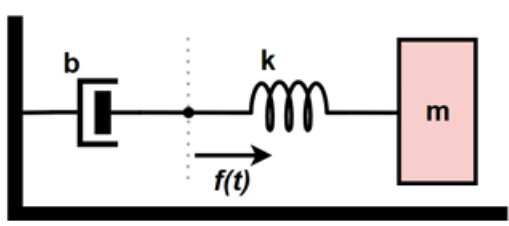


Task 5 (EE&ME): Perform a simulation for this analytical solution in Python/MATLAB, to validate the analogy between these two systems.

- By looking at the simulations that were made using Python and LTspice (*files are attached*), we can see that the mechanical displacement of the mass and the changes in voltage across the capacitor has identical oscillation and decay patterns which can be explained by second order differential equations.



Part 2: Mechanical to Electrical (Force-Current) Conversion

Mechanical Design (MSD)	Equivalent Electrical Design (RLC)
 The diagram shows a mechanical system. A mass m (represented by a pink rectangle) is connected to a spring k (represented by a coiled line) and a damper b (represented by a rectangle with a diagonal line). The damper is connected to a fixed wall on the left. An external force $f(t)$ (represented by an arrow) is applied to the mass to the right. A vertical dashed line indicates the equilibrium position.	

Task 1 (ME):

- $m \rightarrow$ mass
- $b \rightarrow$ damping coefficient
- $k \rightarrow$ spring constant
- $f(t) \rightarrow$ external force
- $x(t) \rightarrow$ position

$$\sum F = m \frac{d^2 x}{dt^2}(t)$$

Forces

- Spring: $-kx(t)$
- Damper: $-b \frac{dx}{dt}(t)$
- External Force: $f(t)$

$$m \frac{d^2 x}{dt^2}(t) + b \frac{dx}{dt}(t) + kx(t) = f(t)$$

Analytical Solution

For a homogeneous solution:

$$m \frac{d^2 x}{dt^2}(t) + b \frac{dx}{dt}(t) = 0$$

Characteristic equation:

$$mr^2 + br + k = 0$$

Roots:

$$r = \frac{(b \pm (b^2 - 4mk)^{0.5})}{2m}$$

1. Overdamped: $b^2 > 4mk$
2. Critical damping: $b^2 = 4mk$
3. Underdamped: $b^2 < 4mk$

Underdamped solution:

$$x(t) = e^{-\frac{b}{2m}t} \cdot (C_1 \cdot \cos(w_d \cdot t) + C_2 \cdot \sin(w_d \cdot t))$$

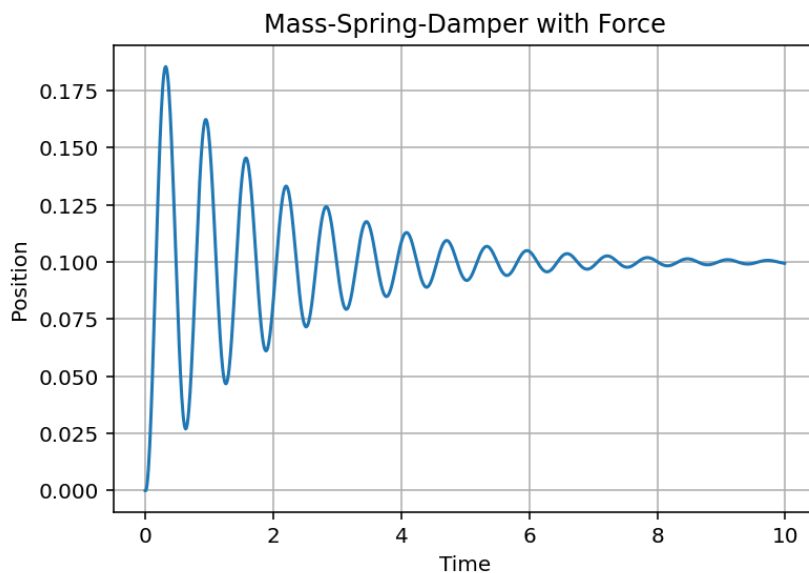
$$w_d = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$$

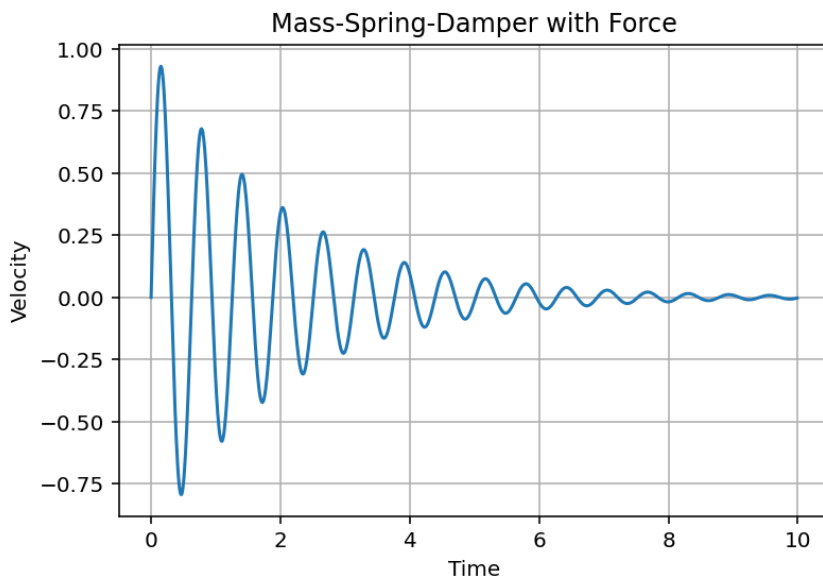
Assumed variable and conditions: $m = 1kg$, $b = 2 \frac{N \cdot s}{m}$, $k = 100 \frac{N}{m}$ and initial conditions

$F_0 = 10N$ and zero replacement at $t = 0$. Here is the solved equation:

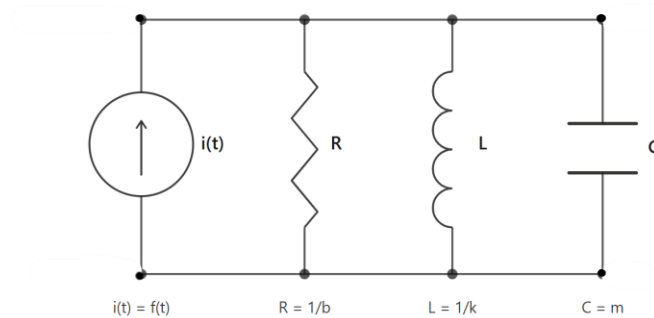
$$x(t) = e^{(-0.25t)}(-0.5 \cos(4.465t) - 0.028 \sin(4.465t)) + 0.5$$

Task 2 (EE-ME): Perform a simulation for this analytical solution in Python/MATLAB.





Task 3 (ME): Draw the electrical circuit that corresponds to this model.



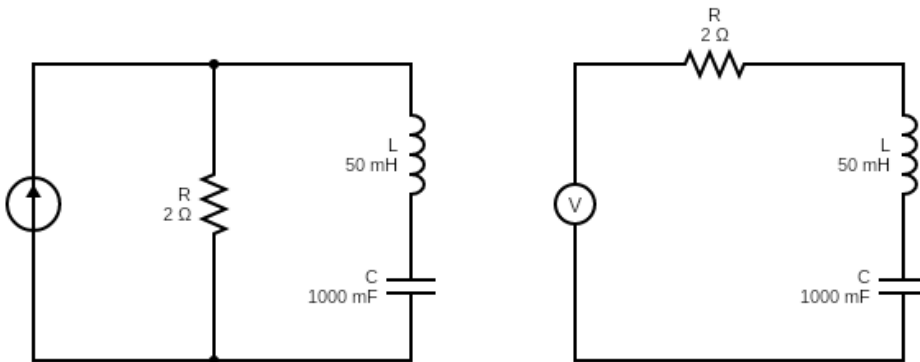
This diagram translates a mechanical Mass-Spring-Damper system into an electrical circuit using the Force-Current (F-I) analogy.

- Parallel Layout: Of this analogy, voltage (V) represents velocity (v). Since all mechanical components move at the same velocity, their electrical counterparts must be in parallel to share the same voltage.
- Component Mapping:
 - Current Source $i(t)$: Represents the applied mechanical force $f(t)$.
 - Capacitor C : Represents the mass m .
 - Resistor R : Represents the damper, where $R = \frac{1}{b}$.
 - Inductor L : Represents the spring, where $L = \frac{1}{k}$.
- Governing Equation: The circuit's node equation Kirchhoff's Current Law:

$$i = i_R + i_L + i_C$$

The exact mathematical equivalent of the mechanical system's force balance Newton's Second Law: $f = f_b + f_k + f_m$

Task 4 (EE): Using Thevenin, Norton, source transformation, simplify the circuit into a single mesh circuit and write down the mesh equation.



$$V_s(t) = V_R + V_L + V_C$$

$$V_s(t) = R \cdot i(t) + L \cdot \frac{di}{dt} + \frac{1}{C} \int i(t) dt$$

$$\frac{f(t)}{b} = \frac{1}{b} i(t) + \frac{1}{k} \frac{di}{dt} + \frac{1}{m} \int i(t) dt$$

Task 5 (EE): Perform an LTSpice simulation to demonstrate the analogy between these two Systems.

In this part, the main goal was to show that the mechanical mass-spring-damper system and the electrical RLC circuit behave in the exact same way. To prove this, we simulated the simplified single-mesh circuit from Task 4 in LTSpice using a transient analysis.

The component values in LTSpice were set based on the force-current analogy we established earlier: $C = m$, $L = 1/k$, $R = 1/b$, and the voltage source $V_s(t) = f(t)/b$. We applied a [step / sine] input to observe how the circuit reacts over time.

When we compare the LTSpice current results with the analytical plot generated in [MATLAB / Python] from Task 2 (files are attached), we can see that the two graphs are identical. The electrical current has the exact same waveform, settling time, and peak values as the mechanical response.

This perfect match clearly shows that the second-order differential equations for both systems are practically the same. It also verifies that our source transformation step was correct and the mechanical system can be accurately analyzed using an equivalent electrical circuit